Stage 8 Maths — Practice Test 1 — Mark Scheme

Paper 1: 60 minutes · no calculator · Paper 2: 60 minutes · calculator allowed · Total 100 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 60 minutes — 50 marks

1 **[1]**

Question 1

Work out $20 - 3 \times (2 + 4)^2 \div 12$

Answer: 11

Notes / working: Brackets first: 2 + 4 = 6. Then the index: $6^2 = 36$. Then × and ÷ from left to right: 3 × 36 = 108, 108 ÷ 12 = 9. Last, the subtraction: 20 − 9 = **11**. Common mistake: working 20 − 3 = 17 first.

2 **[1]**

Question 2

Which of these numbers is an integer but **not** a natural number?

- ☐ A) −7
- ☐ B) 7
- ☐ C) −7/2
- ☐ D) 0.7

Answer: A) −7

Notes / working: Natural numbers are the counting numbers 1, 2, 3, … Integers also include 0 and the negative whole numbers. −7 is a whole number, so it is an integer, but it is negative, so it is not a natural number. 7 is both; −7/2 and 0.7 are rational numbers but not integers. Answer **A (−7)**.

3 **[1]**

Question 3

Hassan writes: $3^4 \times 3^2 = 9^6$

Explain Hassan's mistake and write the correct answer as a single power of 3.

Answer: 1 mark: correct answer 3^6 AND a correct explanation, e.g. 'the base stays 3, you only add the indices', 'he multiplied the 3s / bases', 'he should not multiply the bases'. Do not accept 3^6 with no explanation, or 3^8 (multiplying the indices).
Notes / working: When you multiply powers of the **same base**, keep the base and **add** the indices: $3^4 \times 3^2 = 3^{4+2} = \mathbf{3^6}$. Hassan wrongly multiplied the bases ($3 \times 3 = 9$). Check: $3^4 \times 3^2 = 81 \times 9 = 729 = 3^6$, but $9^6 = 531\,441$.

4 **[1]**

(a) Work out $\sqrt[3]{(-64)}$

Answer: -4
Notes / working: $(-4) \times (-4) \times (-4) = 16 \times (-4) = -64$, so $\sqrt[3]{(-64)} = \mathbf{-4}$. A negative number has a negative cube root.

(b) Work out $(-3)^2 + \sqrt[3]{(-8)}$ **[1]**

Answer: 7
Notes / working: $(-3)^2 = (-3) \times (-3) = 9$ (a negative number squared is positive). $\sqrt[3]{(-8)} = -2$ because $(-2)^3 = -8$. So $9 + (-2) = \mathbf{7}$. Common mistake: $(-3)^2 = -9$.

5 **[1]**

(a) Work out 34×0.1

Answer: 3.4
Notes / working: Multiplying by 0.1 is the same as dividing by 10, so the digits move one place to the right: $34 \times 0.1 = \mathbf{3.4}$.

(b) Work out $2.7 \div 0.01$ **[1]**

Answer: 270
Notes / working: Dividing by 0.01 is the same as multiplying by 100 (there are 100 hundredths in 1), so $2.7 \div 0.01 = \mathbf{270}$. Common mistake: 0.027 (dividing by 100 instead).

6 **[1]**

Question 6
Which fraction is equal to a **recurring** decimal?

- ☐ A) 3/8
- ☐ B) 7/20
- ☐ C) 5/6
- ☐ D) 9/25

Answer: C) 5/6
Notes / working: 3/8 = 0.375, 7/20 = 0.35 and 9/25 = 0.36 all stop (terminate). 5/6 = 0.8333… with the 3 repeating for ever. A fraction in its simplest form terminates only when the denominator has no prime factors other than 2 and 5; 6 = 2 × 3 has a factor 3, so 5/6 recurs. Answer **C**.

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7 **[2]**

Question 7
Work out 4 1/5 − 1 2/3
Show your working. Give your answer as a mixed number in its simplest form.

Answer: 1 mark: correct method with a common denominator of 15, e.g. 63/15 − 25/15, or 3 + 3/15 − 10/15, or 4 3/15 − 1 10/15 (oe). 1 mark: 2 8/15 (cao). Accept 38/15 for the method mark only.
Notes / working: Estimate first: about 4 − 1.5 = 2.5. Change to improper fractions: 4 1/5 = 21/5 = 63/15 and 1 2/3 = 5/3 = 25/15. Then 63/15 − 25/15 = 38/15 = **2 8/15**. Or: whole numbers 4 − 1 = 3, then 1/5 − 2/3 = 3/15 − 10/15 = −7/15, so 3 − 7/15 = 2 8/15. 2 8/15 ≈ 2.5, which matches the estimate.

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8 A bakery in Heliopolis makes biscuits and bread. **[1]**

(a) The baker has 84 chocolate biscuits and 120 date biscuits. She puts all of them into bags. Every bag has the same number of chocolate biscuits and the same number of date biscuits, and none are left over.
What is the **greatest** number of bags she can fill?

Answer: 12
Notes / working: This is the highest common factor of 84 and 120. 84 = 2^2 × 3 × 7 and 120 = 2^3 × 3 × 5. Common prime factors: 2^2 × 3 = **12**. Check: 84 ÷ 12 = 7 and 120 ÷ 12 = 10, both whole numbers.

(b) How many date biscuits are in each bag? **[1]**

Answer: 10
Notes / working: 120 ÷ 12 = **10** date biscuits (and 7 chocolate biscuits) in each bag. Follow through: 120 ÷ your answer to (a).

(c) One batch of bread dough uses 2 3/4 kg of flour. **[2]**
Work out how much flour the baker needs for 6 batches. Show your working.

> **Answer:** 16.5 or 33/2 kg
> **Notes / working:** Estimate: 3 × 6 = 18 kg, so a bit less than 18. Method: 2 3/4 × 6 = (2 × 6) + (3/4 × 6) = 12 + 18/4 = 12 + 4 1/2 = **16 1/2 kg**. Or 11/4 × 6 = 66/4 = 16 1/2. Marks: 1 mark for a correct method (e.g. 12 + 18/4 or 66/4 seen); 1 mark for 16 1/2 or 16.5.

(d) The baker has 6 kg of sesame seeds. She puts them into bags that each hold 3/8 kg. **[1]**
How many bags can she fill?

> **Answer:** 16
> **Notes / working:** 6 ÷ 3/8 = 6 × 8/3 = 48/3 = **16** bags. Check: 16 × 3/8 = 48/8 = 6 kg.

(e) She has 13.2 kg of biscuit dough. Each baking tray holds 0.4 kg of dough. **[2]**
How many trays can she fill? Show your working.

> **Answer:** 33
> **Notes / working:** 13.2 ÷ 0.4: multiply both numbers by 10 so you divide by a whole number: 132 ÷ 4 = **33** trays. Estimate check: 13 ÷ 0.4 is about 32. Marks: 1 mark for 132 ÷ 4 (or an equivalent method); 1 mark for 33.

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9 **[1]**

Question 9
Work out 0.25 × 37 × 4 without a calculator.
Show how you can change the order to make the calculation easier.

> **Answer:** 1 mark: answer 37 with a method that groups 0.25 × 4 = 1 (e.g. 0.25 × 4 × 37 = 1 × 37). Do not accept 37 with no working.
> **Notes / working:** Multiplication can be done in any order (the commutative law): 0.25 × 37 × 4 = 0.25 × 4 × 37 = 1 × 37 = **37**.

10 **[1]**

Question 10
Which symbol makes this statement true?
−0.65 ☐ −2/3

- ☐ A) <
- ☐ B) >
- ☐ C) =
- ☐ D) ≤

Answer: B) >
Notes / working: −2/3 = −0.666… . On a number line −0.65 is to the right of −0.666…, so it is the larger number: −0.65 **>** −2/3. Answer **B**. Common mistake: thinking −0.666… is bigger because 0.666 > 0.65 — with negative numbers the order is reversed.

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11 Year 8 students from a school in Cairo go on a trip to Luxor. **[1]**

(a) The total cost of the trip is E£48 652.
Round E£48 652 to 2 significant figures.

Answer: 49000
Notes / working: The first two significant figures are 4 and 8 (48 thousand). The next digit is 6, which is 5 or more, so round up: **49 000**. Common mistake: 48 or 49 (you must keep the place value).

(b) The ratio of students to teachers on the trip is 15 : 2. There are 68 people altogether. **[1]**
How many teachers are on the trip?

Answer: 8
Notes / working: 15 + 2 = 17 parts. 68 ÷ 17 = 4 people in each part. Teachers = 2 × 4 = **8** (students = 15 × 4 = 60; 60 + 8 = 68 ✓).

(c) The group spends 45 minutes at a museum and 2 hours at the Karnak temples. **[1]**
Write the ratio 45 minutes : 2 hours in its simplest form.

Answer: 3:8
Notes / working: Use the same units first: 2 hours = 120 minutes. 45 : 120. The HCF of 45 and 120 is 15, so divide both by 15: **3 : 8**. Common mistake: 45 : 2 (different units).

(d) The school shares E£4500 between a boat trip, museum tickets and lunch in the ratio 2 : 3 : 5. **[2]**

How much money is for lunch? Show your working.

> **Answer:** 2250
> **Notes / working:** 2 + 3 + 5 = 10 parts. One part = 4500 ÷ 10 = E£450. Lunch = 5 × 450 = **E£2250**. Check: 900 + 1350 + 2250 = 4500 ✓. Marks: 1 mark for 4500 ÷ 10 or 450 seen; 1 mark for 2250.

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12 **[1]**

Question 12

Which of these is a **formula**?

- ☐ A) 3x + 7
- ☐ B) 2x + 5 = 11
- ☐ C) 4n − 1 > 6
- ☐ D) P = 2l + 2w

> **Answer:** D) P = 2l + 2w
> **Notes / working:** A formula connects two or more letters (variables) that stand for quantities, so you can work out one from the others: P = 2l + 2w gives the perimeter of a rectangle. 3x + 7 is an expression (no equals sign); 2x + 5 = 11 is an equation (true for one value, x = 3); 4n − 1 > 6 is an inequality. Answer **D**.

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13 **[1]**

(a) Expand 4x(3x − 2)

> **Answer:** 12x^2-8x or 12x^2-8x or -8x+12x^2
> **Notes / working:** Multiply each term in the bracket by 4x: 4x × 3x = 12x^2 and 4x × (−2) = −8x. Answer **12x^2 − 8x**. Common mistake: 12x − 8x or 12x^2 − 2.

(b) Factorise completely 8a^2 + 12a **[2]**

> **Answer:** 2 marks: 4a(2a + 3). 1 mark: a correct partial factorisation, e.g. 2a(4a + 6), 4(2a^2 + 3a) or a(8a + 12).
> **Notes / working:** The HCF of 8a^2 and 12a: numbers HCF(8, 12) = 4; letters a^2 and a have a in common. So the HCF is 4a. 8a^2 ÷ 4a = 2a and 12a ÷ 4a = 3. Answer **4a(2a + 3)**. Check by expanding: 4a × 2a + 4a × 3 = 8a^2 + 12a ✓. 'Completely' means nothing more can be taken out of the bracket.

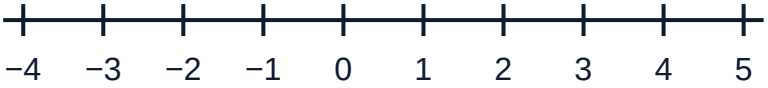
14 **[2]**

Question 14
Solve 5x − 7 = 2x + 11
Show your working.

Answer: 1 mark: a correct first step collecting terms, e.g. 3x − 7 = 11 or 3x = 18 or 5x = 2x + 18. 1 mark: x = 6.
Notes / working: Subtract 2x from both sides: 3x − 7 = 11. Add 7 to both sides: 3x = 18. Divide by 3: **x = 6**. Check: 5 × 6 − 7 = 23 and 2 × 6 + 11 = 23 ✓.

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15 **[1]**



Question 15
Show the inequality −2 < x ≤ 3 on the number line. (Copy the number line onto paper.)

Answer: 1 mark: open (empty) circle at −2, closed (filled) circle at 3, and a line joining them. Do not award if the circles are the wrong way round.
Notes / working: x is **greater than** −2 (−2 itself is not included) → open circle at −2. x is **less than or equal to** 3 (3 is included) → filled circle at 3. Join the circles with a straight line.

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16 Here is a sequence: 7, 4, 1, −2, … **[2]**

(a) Find an expression for the nth term of the sequence.

Answer: 10-3n or -3n+10 or 10-3×n or -3×n+10
Notes / working: The terms go down by 3 each time, so the nth term starts with −3n. When n = 1, −3n = −3, and you need 7, so add 10: **10 − 3n** (or −3n + 10). Check: n = 2 gives 10 − 6 = 4 ✓.
Marks: 1 mark for −3n + c (any c, or 'goes down by 3' used correctly); 2 marks for 10 − 3n.

(b) Is −50 a term in this sequence? Show how you decide. **[1]**

Answer: 1 mark: 'Yes' with a correct reason, e.g. solving 10 − 3n = −50 to get n = 20 (a whole number), or saying −50 is the 20th term. Follow through from an incorrect nth term in (a) if the method is correct.
Notes / working: Solve 10 − 3n = −50: −3n = −60, so n = 20. n is a whole number, so **yes**, −50 is the 20th term. (If n had come out as a decimal, the number would not be in the sequence.)

17 **[1]**

Question 17

Here is a function machine.

input → × 4 → − 3 → output

The output is 25. Work out the input.

Answer: 7

Notes / working: Use the inverse operations in reverse order: 25 + 3 = 28, then 28 ÷ 4 = **7**. Check: 7 × 4 − 3 = 25 ✓. Common mistake: 25 ÷ 4 + 3.

18 **[1]**

Question 18

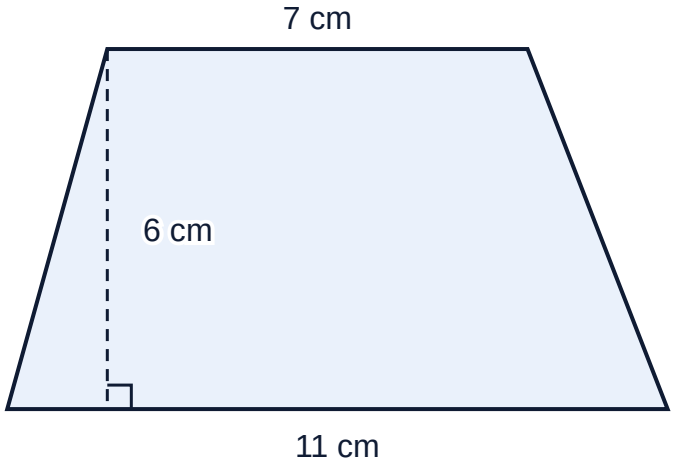
Nour says, "A rhombus is always a square."

Is Nour correct? Explain your answer.

Answer: 1 mark: 'No' with a correct reason, e.g. a rhombus does not have to have right angles; a square is a special rhombus (but not the other way round); a counter-example such as a rhombus with angles 60° and 120°. Do not accept 'No' with no reason.

Notes / working: No. A rhombus has four equal sides, but its angles do not have to be 90°. A square has four equal sides **and** four right angles, so every square is a rhombus, but not every rhombus is a square. One counter-example (a rhombus with angles 70° and 110°) is enough to show Nour is wrong.

19 **[2]**



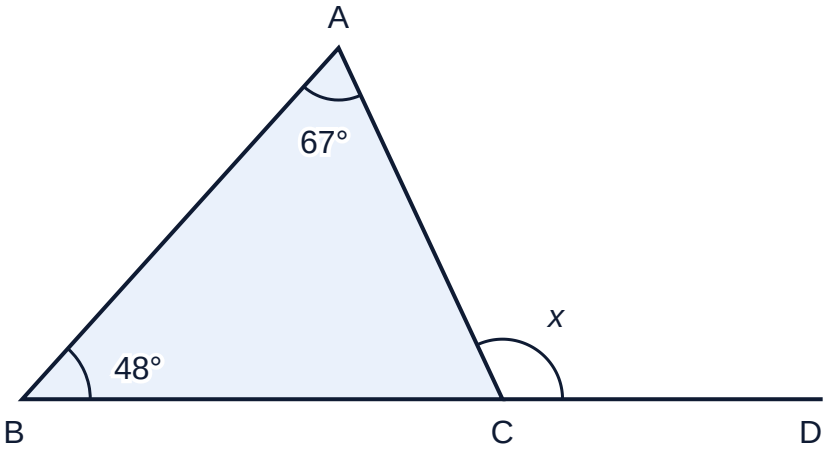
Question 19

Work out the area of this trapezium. Show your working.

Answer: 54 cm^2

Notes / working: Area of a trapezium = ½ × (a + b) × h, where a and b are the parallel sides. ½ × (7 + 11) × 6 = ½ × 18 × 6 = 9 × 6 = **54 cm^2**. Marks: 1 mark for ½ × (7 + 11) × 6 or 18 × 3 seen; 1 mark for 54. Common mistake: using the sloping side, or forgetting the ½ (108).

20 BC is extended to D. **[1]**



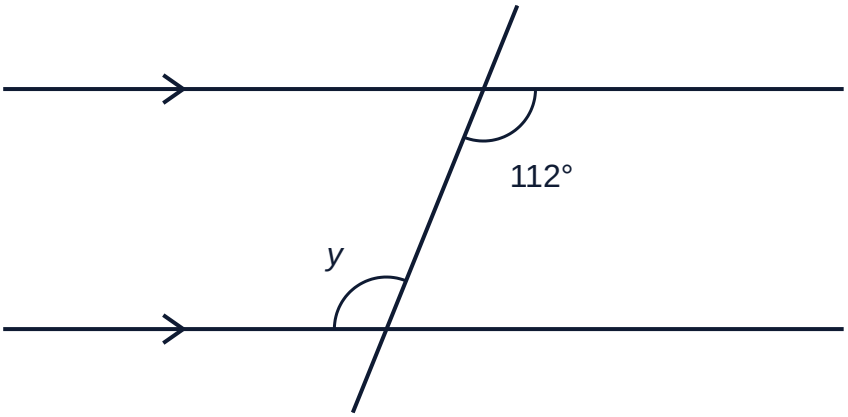
(a) Work out the size of angle x.

Answer: 115 °

Notes / working: The exterior angle of a triangle equals the sum of the two interior opposite angles: x = 48° + 67° = **115°**. (Or: angle ACB = 180 − 48 − 67 = 65°, then x = 180 − 65 = 115°.)

21 (continued) **[1]**

The diagram shows two parallel lines crossed by a straight line.



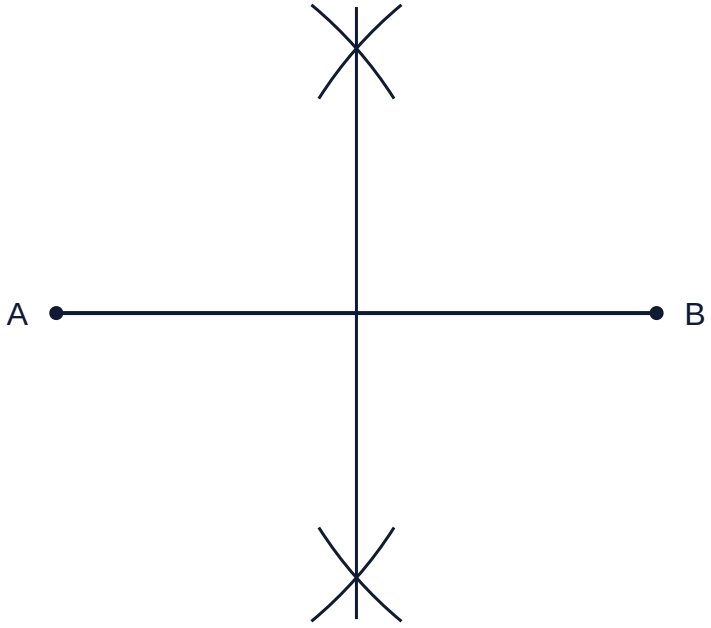
(b) Work out the size of angle y.

Answer: 112 °
Notes / working: Angle y and the 112° angle are between the parallel lines on opposite sides of the crossing line — they are alternate angles, so y = **112°**.

(c) Give a geometric reason for your answer to part (b). **[1]**

Answer: 1 mark: 'alternate angles (are equal)'. The word 'alternate' is needed: do not accept 'Z-angles' alone, 'corresponding' or 'opposite'.
Notes / working: Alternate angles are equal (when the lines are parallel). Use the correct geometric word in a Checkpoint answer — 'alternate', not 'Z-angle'.

22 **[1]**



Question 21

Amir uses a ruler and compasses. He draws arcs with the same radius from A and from B, then a straight line through the points where the arcs cross.

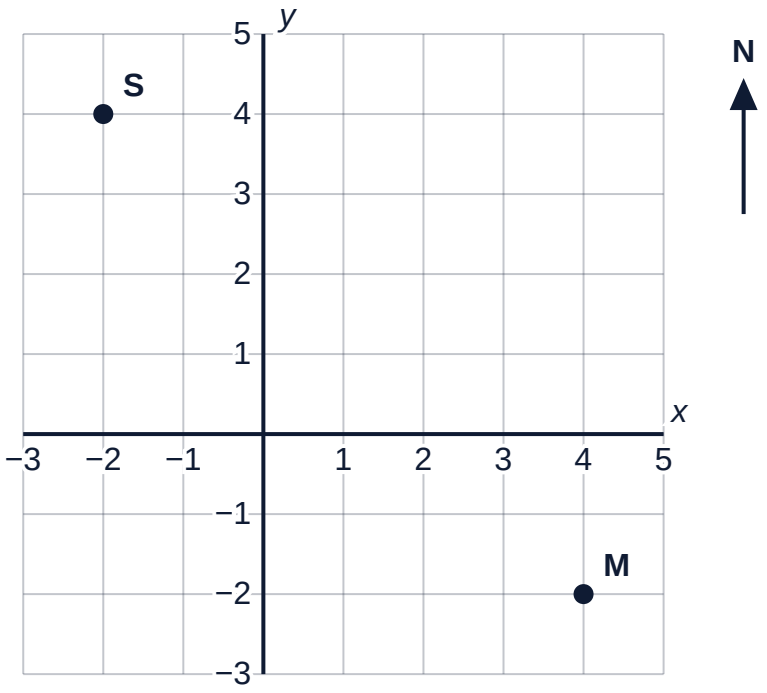
What has he constructed?

- ☐ A) The perpendicular bisector of AB
- ☐ B) The bisector of an angle
- ☐ C) A triangle with sides 5 cm, 5 cm and 3 cm
- ☐ D) A line parallel to AB

Answer: A) The perpendicular bisector of AB

Notes / working: Equal arcs from both ends of a line segment meet at two points that are the same distance from A and from B. The line through them cuts AB in half at right angles: the **perpendicular bisector** of AB. Answer **A**.

23 The map of part of a city is drawn on a coordinate grid. S is a bus station and M is a museum. **[1]**



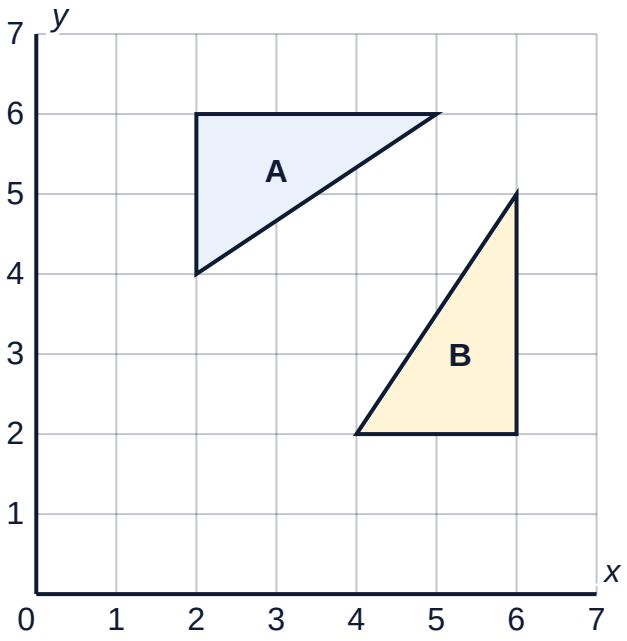
(a) Write down the coordinates of the midpoint of SM.

Answer: (1,1) or 1,1
Notes / working: S is (−2, 4) and M is (4, −2). Midpoint = ((−2 + 4) ÷ 2, (4 + (−2)) ÷ 2) = (2 ÷ 2, 2 ÷ 2) = **(1, 1)**.

(b) The bearing of M from S is 135°. **[1]**
Work out the bearing of S from M.

Answer: 315 °
Notes / working: A back bearing differs by 180°: 135° + 180° = **315°**. Check on the grid: from M, S is up and to the left (north-west), and north-west is 315°.

24 **[2]**



Question 23

Describe fully the single transformation that maps triangle A onto triangle B.

Answer: 1 mark: reflection. 1 mark: in the line y = x. Do not award the second mark for 'diagonal line' or 'y = −x'. A description using two transformations scores 0.
Notes / working: Each vertex (x, y) of A goes to (y, x): (2, 4) → (4, 2), (2, 6) → (6, 2), (5, 6) → (6, 5). The shape is flipped over, so it is a **reflection in the line y = x**. 'Describe fully' needs both the type of transformation and the mirror line.

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25 For the science fair, Salma grew 15 bean plants. The stem-and-leaf diagram shows their heights in cm. **[1]**

Stem	Leaf
1	4 7 8
2	0 3 3 6 9
3	1 2 5 8
4	0 2 6
Key: 2	3 means 23 cm

(a) Find the median height.

Answer: 29 cm
Notes / working: There are 15 heights, so the median is the (15 + 1) ÷ 2 = 8th value. Counting in order: 14, 17, 18, 20, 23, 23, 26, **29** … The median is **29 cm**.

(b) Find the range of the heights. **[1]**

> **Answer:** 32 cm
> **Notes / working:** Range = largest − smallest = 46 − 14 = **32 cm**.

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26 At the science fair there is a game with spinners. **[1]**

(a) The probability that a spinner lands on red is 0.35.
Work out the probability that it does **not** land on red.

> **Answer:** 0.65 or 13/20
> **Notes / working:** 'Red' and 'not red' are complementary events, so their probabilities add up to 1: 1 − 0.35 = **0.65**.

(b) Another spinner has three equal sections numbered 1, 2 and 3. Ali spins it and also flips a fair coin. **[1]**
Work out the probability that he gets an odd number **and** a head.

> **Answer:** 1/3
> **Notes / working:** List all equally likely outcomes: 1H, 1T, 2H, 2T, 3H, 3T — 6 outcomes. Odd and head: 1H and 3H — 2 outcomes. Probability = 2/6 = **1/3**.

Paper 2 — 60 minutes — 50 marks

1 A mobile-phone shop in Nasr City sells phones and accessories. **[2]**

(a) A phone costs E£14 500. In a sale the price is reduced by 12%.
Work out the sale price. Show your working.

Answer: 12760
Notes / working: Method 1: 12% of 14 500 = 0.12 × 14 500 = 1740, then 14 500 − 1740 = **E£12 760**. Method 2 (multiplier): 100% − 12% = 88%, so 14 500 × 0.88 = 12 760. Marks: 1 mark for 1740 or 14 500 × 0.88 seen; 1 mark for 12 760.

(b) The price of another phone increased from E£9600 to E£11 040. **[1]**
(i) Work out the absolute change in the price.

Answer: 1440
Notes / working: Absolute change = new − old = 11 040 − 9600 = **E£1440**. The absolute change is an amount of money, not a percentage.

(b)(ii) Work out the percentage increase. Show your working. **[2]**

Answer: 15 %
Notes / working: Percentage increase = change ÷ original × 100 = 1440 ÷ 9600 × 100 = **15%**. Marks: 1 mark for 1440 ÷ 9600 (or 11 040 ÷ 9600 = 1.15) seen; 1 mark for 15. Common mistake: dividing by the new price (1440 ÷ 11 040 = 13.0%).

(c) 3 screen protectors cost E£247.50. **[1]**
Work out the cost of 7 screen protectors.

Answer: 577.5
Notes / working: The cost is in direct proportion to the number bought. One costs 247.50 ÷ 3 = E£82.50. Seven cost 7 × 82.50 = **E£577.50**.

2 **[1]**

(a) Estimate the value of 487 × 62 ÷ 19
Show your rounding.

Answer: 1 mark: 500 × 60 ÷ 20 (each number rounded to 1 significant figure) and 1500. Accept other sensible roundings giving a correct value for them, e.g. 500 × 60 ÷ 19 → about 1600, or 490 × 60 ÷ 20 = 1470. Do not accept the exact answer without rounding.
Notes / working: Round each number to 1 significant figure: 487 ≈ 500, 62 ≈ 60, 19 ≈ 20. Then 500 × 60 = 30 000 and 30 000 ÷ 20 = **1500**.

(b) Use your calculator to work out 487 × 62 ÷ 19. Give your answer correct to 3 significant figures. **[1]**

> **Answer:** 1590
> **Notes / working:** 487 × 62 = 30 194, and 30 194 ÷ 19 = 1589.157… . To 3 significant figures: the fourth figure is 9, so round up: **1590**. This is close to the estimate of 1500, so the answer is sensible.

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3 Six teams play in a school football tournament. Each team plays every other team once. **[1]**

Team	Won	Drawn	Lost	Goals scored	Goals conceded	Points
Al-Nour	4	1	0	15	4	13
Falcons	3	1	1	9	7	10
Pharaohs	2	2	1	12	8	?
Nile Stars	2	1	2	9	10	7
Delta	1	1	3	7	12	4
Sphinx	0	0	5	5	16	0

The number of points P is given by the formula P = 3w + d, where w is the number of wins and d is the number of draws.

(a) Work out the number of points for the Pharaohs.

> **Answer:** 8
> **Notes / working:** P = 3w + d = 3 × 2 + 2 = 6 + 2 = **8** points.

(b) Rearrange the formula P = 3w + d to make w the subject. **[2]**

> **Answer:** 1 mark: a correct first step, e.g. P − d = 3w. 1 mark: w = (P − d)/3 or w = (P − d) ÷ 3 (oe, e.g. w = P/3 − d/3). Do not accept w = P − d/3 without brackets.
> **Notes / working:** Subtract d from both sides: P − d = 3w. Divide both sides by 3: **w = (P − d)/3**. Check with Al-Nour: (13 − 1) ÷ 3 = 4 wins ✓.

(c) Work out the mean number of goals scored per team. **[1]**

> **Answer:** 9.5
> **Notes / working:** Total goals scored = 15 + 9 + 12 + 9 + 7 + 5 = 57. Mean = 57 ÷ 6 = **9.5**.

(d) Find the median number of goals scored. **[1]**

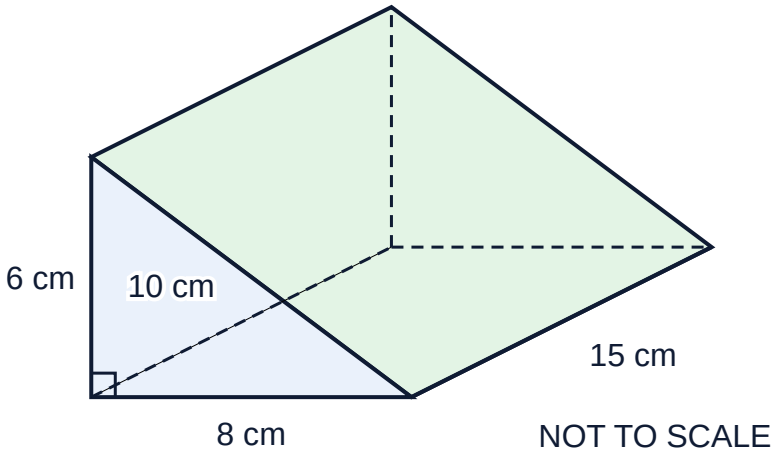
> **Answer:** 9
> **Notes / working:** In order: 5, 7, 9, 9, 12, 15. There are 6 values, so the median is halfway between the 3rd and 4th: (9 + 9) ÷ 2 = **9**.

4 The bakery from Paper 1 uses trays and boxes. **[2]**

(a) A round baking tray has a diameter of 38 cm.
Work out the circumference of the tray. Give your answer correct to 1 decimal place.

Answer: 119.4 cm
Notes / working: Circumference = π × diameter = π × 38 = 119.380… = **119.4 cm** (1 d.p.). Marks: 1 mark for π × 38 seen (or 2 × π × 19); 1 mark for 119.4. Using π = 3.14 gives 119.32, which earns the method mark only. Common mistake: π × 19 (using the radius).

[2]



(b) A slice of cake is a triangular prism. Its cross-section is a right-angled triangle with sides 6 cm, 8 cm and 10 cm, and the slice is 15 cm long.
Work out the volume of the slice. Show your working.

Answer: 360 cm^3
Notes / working: Volume of a prism = area of cross-section × length. The cross-section is a right-angled triangle: ½ × 8 × 6 = 24 cm^2. Volume = 24 × 15 = **360 cm^3**. Marks: 1 mark for 24 (area of triangle) seen; 1 mark for 360. Common mistake: 8 × 6 × 15 = 720 (forgetting the ½).

(c) Work out the total surface area of the slice. Show your working. **[2]**

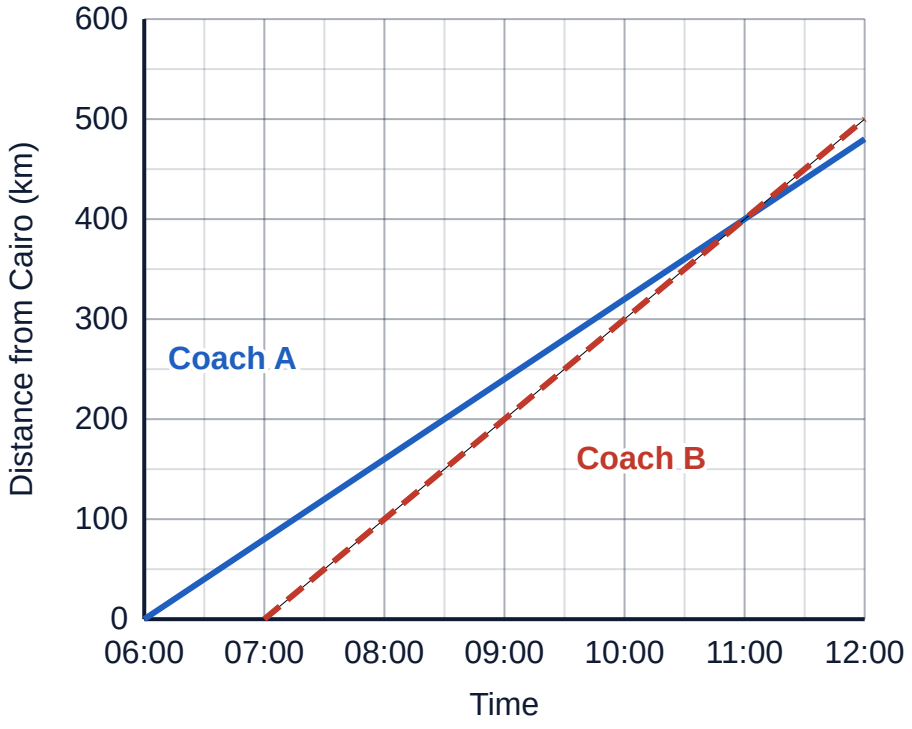
Answer: 408 cm^2
Notes / working: Two triangles: 2 × 24 = 48 cm^2. Three rectangles: 8 × 15 = 120, 6 × 15 = 90 and 10 × 15 = 150, total 360 cm^2. Surface area = 48 + 360 = **408 cm^2**. Marks: 1 mark for at least three of the five face areas correct (24, 24, 120, 90, 150); 1 mark for 408.

5 This question is about the Year 8 trip to Luxor. **[1]**

(a) The coach travels 660 km from Cairo to Luxor.
1 mile = 1.6 km. Convert 660 km to miles.

Answer: 412.5 miles
Notes / working: km → miles: divide by 1.6. 660 ÷ 1.6 = **412.5 miles**. (Or use 1 km ≈ 5/8 mile: 660 × 5/8 = 412.5.) Check: a distance in miles is a smaller number than the same distance in km.

6 This question is about the Year 8 trip to Luxor. **[1]**
Two coaches leave Cairo for Luxor. The graph shows their journeys from 06:00 to 12:00.



(b)(i) At what time does Coach B pass Coach A?

Answer: 11:00 or 11.00 or 11am or 11 am or 1100 or 11:00am or 11:00 am or 11 o'clock or 11
Notes / working: The coaches are at the same place at the same time where the two lines **cross**. The lines cross at **11:00**.

(b)(ii) How far from Cairo are the coaches when this happens? **[1]**

Answer: 400 km
Notes / working: At the crossing point, read across to the distance axis: **400 km**. (Check: Coach A at 80 km per hour for 5 hours = 400 km; Coach B at 100 km per hour for 4 hours = 400 km.)

(b)(iii) Which coach travels faster? Explain how the graph shows this. **[1]**

> **Answer:** 1 mark: Coach B, with a reason referring to the graph, e.g. its line is steeper; it travels 100 km each hour but A travels 80 km each hour; it starts later but catches up. Do not accept 'Coach B' alone.
> **Notes / working: Coach B.** Its line is steeper: it covers 100 km every hour (500 km from 07:00 to 12:00), while Coach A covers 80 km every hour (480 km in 6 hours). On a distance–time graph a steeper line means a greater speed.

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7 This question is about the Year 8 trip to Luxor. **[1]**
A hotel in Luxor charges a booking fee of E£300 plus E£450 for each night.

(c) Write an equation for the total cost y (in E£) for x nights.

> **Answer:** 1 mark: y = 450x + 300 (oe, e.g. y = 300 + 450x). Do not accept an expression without 'y =', or y = 300x + 450.
> **Notes / working:** Each night adds E£450, so the cost goes up 450 for each x; the fixed fee is 300. **y = 450x + 300**. This is a straight line with gradient 450 and y-intercept 300.

(d) The total cost for one teacher's stay is E£2550. **[1]**
How many nights did the teacher stay?

> **Answer:** 5
> **Notes / working:** 450x + 300 = 2550 → 450x = 2250 → x = 2250 ÷ 450 = **5** nights. Check: 450 × 5 + 300 = 2550 ✓.

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8 Here is a table of values for y = 3x − 2. **[1]**

x	−1	0	1	2	3
y	?	−2	1	?	7

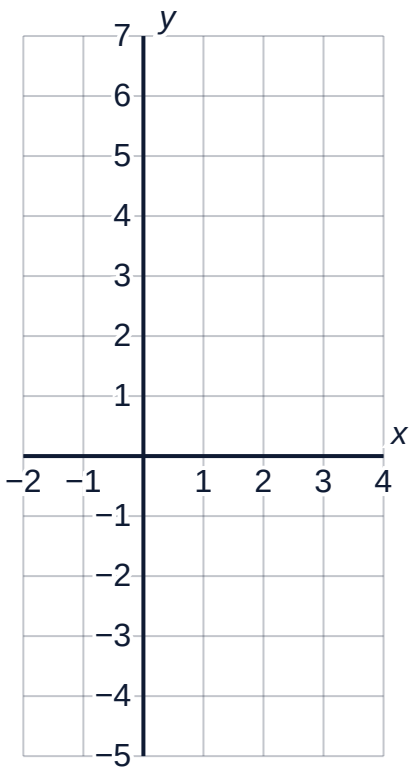
(a) Work out the value of y when x = −1.

> **Answer:** -5
> **Notes / working:** y = 3 × (−1) − 2 = −3 − 2 = **−5**.

(b) Work out the value of y when x = 2. **[1]**

> **Answer:** 4
> **Notes / working:** y = 3 × 2 − 2 = 6 − 2 = **4**. Check the pattern: the y-values go up by 3 each time (−5, −2, 1, 4, 7).

[1]



(c) Draw the graph of y = 3x − 2 for values of x from −1 to 3. (Copy the grid onto squared paper.)

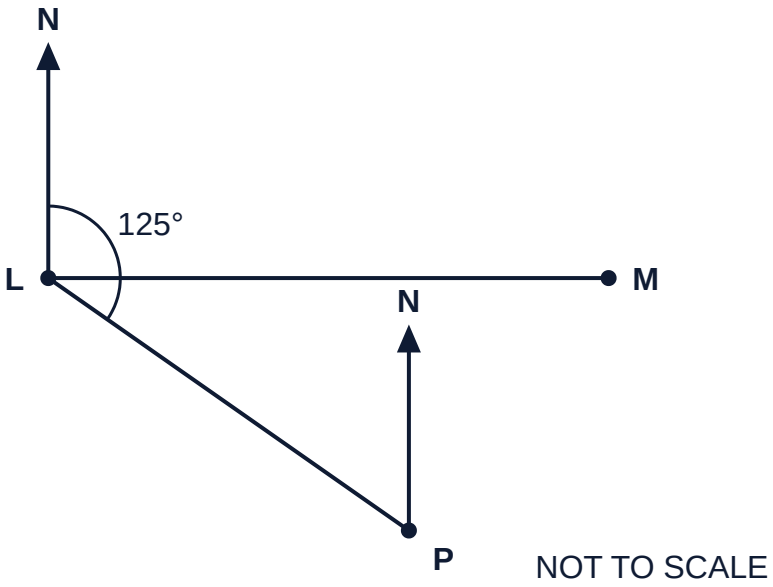
Answer: 1 mark: the five points (−1, −5), (0, −2), (1, 1), (2, 4), (3, 7) plotted correctly and joined with one straight ruled line from x = −1 to x = 3. Follow through: if one value in your table (part b) was wrong, award the mark if every point from your table is plotted correctly and the four correct points are joined by a ruled straight line.
Notes / working: Plot (−1, −5), (0, −2), (1, 1), (2, 4) and (3, 7). They lie on a straight line: join them with a ruler from x = −1 to x = 3. The line crosses the y-axis at −2 (the y-intercept) and goes up 3 for every 1 across (gradient 3).

(d) Which line is **parallel** to y = 3x − 2? **[1]**

- ☐ A) y = 2x − 3
- ☐ B) y = 3x + 4
- ☐ C) y = −3x − 2
- ☐ D) y = x − 2

Answer: B) y = 3x + 4
Notes / working: Parallel lines have the same gradient. In y = mx + c the gradient is m, so we need m = 3: **y = 3x + 4** (answer **B**). y = −3x − 2 has gradient −3 (it slopes the other way) and y = x − 2 only shares the y-intercept.

9 The map shows three places in a city: a library L, a mosque M and a park P. **[1]**
M is due east of L. The bearing of P from L is 125°.



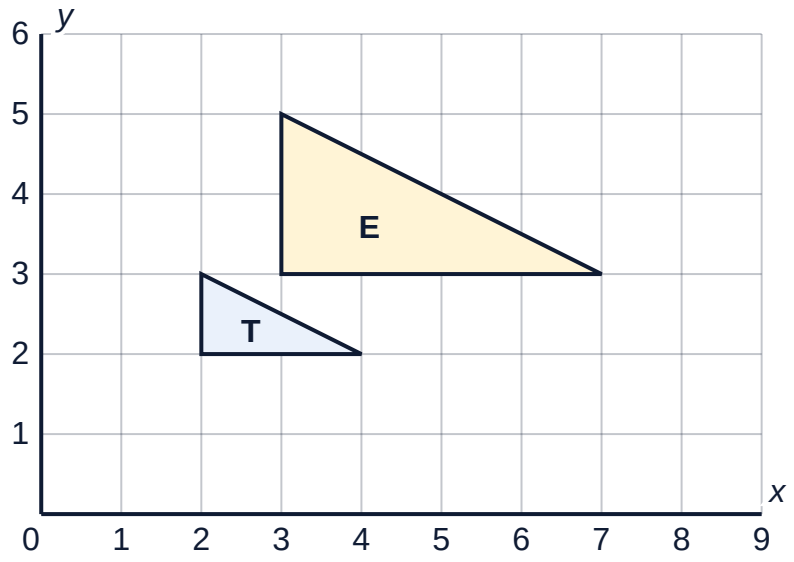
(a) Work out the bearing of L from P.

Answer: 305 °
Notes / working: The back bearing is 125° + 180° = **305°**. At P, measure clockwise from north all the way round to the line PL.

(b) Work out the size of angle MLP. **[1]**

Answer: 35 °
Notes / working: Due east is a bearing of 090°. Angle MLP = 125° − 90° = **35°**.

10 Triangle T has vertices (2, 2), (4, 2) and (2, 3). Triangle E is an enlargement of triangle T. **[1]**



(a) Write down the scale factor of the enlargement.

Answer: 2
Notes / working: The base of T is 2 squares long; the base of E is 4 squares long. 4 ÷ 2 = **2**. The height also doubles (1 → 2).

(b) Find the coordinates of the centre of enlargement. **[1]**

Answer: (1,1) or 1,1
Notes / working: Draw straight lines from each vertex of E back through the matching vertex of T: (7, 3) through (4, 2), (3, 5) through (2, 3), (3, 3) through (2, 2). They meet at **(1, 1)**. Check: from (1, 1) to (4, 2) is 3 right, 1 up; doubled is 6 right, 2 up → (7, 3) ✓.

[1]

$$\text{vector} \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

(c) Triangle T is translated by the vector shown.
Write down the coordinates of the three vertices of the image.

Answer: 1 mark: (6, 0), (8, 0) and (6, 1), all three correct. Do not award if any vertex is wrong.
Notes / working: The top number of the vector is the movement in x (4 to the right); the bottom number is the movement in y (2 down). Add 4 to each x and subtract 2 from each y: (2, 2) → (6, 0), (4, 2) → (8, 0), (2, 3) → (6, 1). The image is congruent to T (same size and shape).

(d) To describe a rotation fully, which information do you need? **[1]**

- ☐ A) The centre and the angle only
- ☐ B) The angle and the direction only
- ☐ C) The centre, the angle and the direction
- ☐ D) The centre, the scale factor and the direction

Answer: C) The centre, the angle and the direction
Notes / working: A rotation needs the **centre** of rotation, the **angle** (e.g. 90°) and the **direction** (clockwise or anticlockwise). A scale factor belongs to enlargements. Answer **C**. (For 180° the direction does not matter, but you still state it or say 'half turn'.)

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11 **[1]**

(a) Ravi says, "A regular polygon with rotational symmetry of order 9 has 9 lines of symmetry and 9 sides."
Is Ravi correct? Explain your answer.

Answer: 1 mark: 'Yes' with a reason, e.g. for a regular polygon the number of sides = number of lines of symmetry = order of rotational symmetry; it is a regular nonagon (9 sides) with 9 lines of symmetry.
Notes / working: Yes. In a regular polygon, the number of sides, the number of lines of symmetry and the order of rotational symmetry are all the same. Order 9 means it is a regular nonagon: 9 sides and 9 lines of symmetry.

(b) A 3D shape has 12 vertices and 18 edges. **[1]**
Use Euler's formula to work out the number of faces.

Answer: 8
Notes / working: Euler's formula: $F + V - E = 2$. $F + 12 - 18 = 2$, so $F - 6 = 2$ and **$F = 8$**. (A hexagonal prism has 12 vertices, 18 edges and 8 faces.)

(c) The plan (top view) of a solid is a circle. The front elevation is a rectangle. **[1]**
Name the solid.

Answer: cylinder or a cylinder
Notes / working: From above you see a circle; from the front you see a rectangle. The solid is a **cylinder**. (A cone would have a triangle as its front elevation; a sphere would show a circle from every side.)

12 120 students visited the school science fair. Each student voted for their favourite project area. **[1]**

	Physics	Chemistry	Biology	Total
Boys	18	15	?	50
Girls	22	?	25	70
Total	40	?	?	120

(a) How many girls voted for Chemistry?

Answer: 23
Notes / working: Girls' row: 70 − 22 − 25 = **23**.

(b) Salma draws a pie chart to show the votes of all 120 students. **[2]**
Work out the angle for Biology. Show your working.

Answer: 126 °
Notes / working: Biology total: boys 50 − 18 − 15 = 17, girls 25, so 17 + 25 = 42 students. Each student is 360° ÷ 120 = 3°. Angle = 42 × 3 = **126°**. Marks: 1 mark for 42 seen (or 42/120 × 360 with their 42); 1 mark for 126.

(c) For her own science project Salma collects data. Which of these is **continuous** data? **[1]**

- ☐ A) The number of students in a class
- ☐ B) Favourite science subject
- ☐ C) Shoe size
- ☐ D) The mass of a plant in grams

Answer: D) The mass of a plant in grams
Notes / working: Continuous data is **measured** and can take any value in a range (e.g. 23.47 g). Mass is measured, so it is continuous — answer **D**. Number of students is discrete (counted), shoe size is discrete (only certain sizes), favourite subject is categorical.

(d) Laila wants to know what all 600 students in the school think of the science fair. She asks the 30 students in her own science club. **[1]**
Give **one** reason why this may not be a good sample.

Answer: 1 mark: a valid reason, e.g. the sample is biased / not representative because science-club students probably like science more than other students; only one group is asked; it is not random; 30 out of 600 is a small sample (accept). Do not accept 'it is unfair' with no explanation.
Notes / working: The science club is not **representative** of the whole school: its members probably like science more than most students, so the results are likely to be **biased**. A better method is a random sample of students from every year group.

13 (continued) **[2]**

(e) Classes 8A and 8B took the same science quiz, marked out of 20.

Class	Mean	Range
8A	14.2	6
8B	12.8	15

Compare the scores of the two classes. Write **two** comparisons.

Answer: 1 mark: comparing averages in context, e.g. 8A did better on average because their mean is higher (14.2 > 12.8). 1 mark: comparing spread in context, e.g. 8A's scores were more consistent / less spread out because their range is smaller (6 < 15). Do not accept just quoting the numbers without saying what they mean.
Notes / working: 1. **Average:** 8A's mean (14.2) is higher than 8B's (12.8), so on average 8A scored better. 2. **Spread:** 8A's range (6) is smaller than 8B's (15), so 8A's scores were more consistent; 8B's scores were more spread out. Always compare one average and one measure of spread, and say what they mean.

14 Omar has a spinner with four equal sections coloured red, blue, green and yellow. He spins it 200 times. It lands on red 62 times. **[1]**

(a) Write down the experimental probability that the spinner lands on red.

Answer: 0.31 or 31/100 or 62/200
Notes / working: Experimental probability = number of times red ÷ number of trials = 62 ÷ 200 = **0.31** (or 31/100).

(b) Omar says, "The spinner is biased towards red." **[1]**
Use the theoretical probability to comment on what Omar says.

Answer: 1 mark: a comment that compares with the theoretical probability 0.25 (or 50 out of 200), e.g. 'If fair, red should come up about 50 times; 62 is a lot more, so Omar could be right' or 'He may be right, but he should do more trials to be sure'. Must mention 0.25, 1/4 or 50.
Notes / working: If the spinner is fair, P(red) = 1/4 = 0.25, so in 200 spins you would expect about 0.25 × 200 = 50 reds. Omar got 62, which is quite a lot more, so the spinner **may be biased** towards red. Experimental results always vary, so to be more sure he should do many more spins.

(c) Omar spins the same spinner 500 times. Use his experimental probability to estimate how many times it lands on red. **[1]**

Answer: 155
Notes / working: Expected number = probability × number of trials = 0.31 × 500 = **155**.

15 **[2]**

Question 12

Solve the equation $(3x - 1)/4 = x - 2$

Show your working.

Answer: 1 mark: correct removal of the fraction, $3x - 1 = 4(x - 2)$ or $3x - 1 = 4x - 8$. 1 mark: $x = 7$.
Notes / working: Multiply both sides by 4: $3x - 1 = 4(x - 2) = 4x - 8$. Subtract $3x$ from both sides: $-1 = x - 8$. Add 8: **$x = 7$**. Check: $(3 \times 7 - 1)/4 = 20/4 = 5$ and $7 - 2 = 5$ ✓.